Customer Service Request Analysis

Importing Libraries

In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Step 1 :- Understand the Dataset

Import Dataset

In [2]:
```
df = pd.read_csv('Customer_Analysis.csv')
```

```
C:\Users\DELL\AppData\Local\Temp\ipykernel_14776\2962882006.py:1: DtypeWarning: Colu
mns (48,49) have mixed types. Specify dtype option on import or set low_memory=Fals
e.
  df = pd.read_csv('Customer_Analysis.csv')
```

Visualize Dataset

In [3]:
```
df.head()
```

Out[3]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Lo
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Stre
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Stre
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Stre

5 rows × 53 columns

Print Columns

In [4]:
```
df.columns
```

Out[4]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
```

Shape Of Dataset

In [5]:
```
df.shape
```

Out[5]: (364558, 53)

Check Null Values

In [6]:
```
df.isnull().sum()
```

Out[6]:

```
Unique Key                            0
Created Date                          0
Closed Date                        2381
Agency                                0
Agency Name                           0
Complaint Type                        0
Descriptor                         6501
Location Type                       133
Incident Zip                       2998
Incident Address                  51699
Street Name                       51699
Cross Street 1                    57188
Cross Street 2                    57805
Intersection Street 1            313438
Intersection Street 2            314046
Address Type                       3252
City                               2997
Landmark                         364183
Facility Type                      2389
Status                                0
Due Date                              3
Resolution Description                0
Resolution Action Updated Date     2402
Community Board                       0
Borough                               0
X Coordinate (State Plane)         4030
Y Coordinate (State Plane)         4030
Park Facility Name                    0
Park Borough                          0
School Name                           0
School Number                         0
School Region                         1
School Code                           1
School Phone Number                   0
School Address                        0
School City                           0
School State                          0
School Zip                            1
School Not Found                      0
School or Citywide Complaint     364558
Vehicle Type                     364558
Taxi Company Borough             364558
Taxi Pick Up Location            364558
Bridge Highway Name              364261
Bridge Highway Direction         364261
Road Ramp                        364296
Bridge Highway Segment           364296
Garage Lot Name                  364558
Ferry Direction                  364557
Ferry Terminal Name              364556
Latitude                           4030
Longitude                          4030
Location                           4030
dtype: int64
```

In [7]:
```
df.duplicated().sum()
```

Out[7]:
```
np.int64(0)
```

Remove Agency Name Column

In [8]:
```
df.drop('Agency Name', axis = 1, inplace = True)
```

In [9]:
```
df['Complaint Type'].isnull().sum()
```

Out[9]:
```
np.int64(0)
```

Filling NA Value of Descriptor Column

In [10]:
```
df['Descriptor'].isnull().sum()
```

Out[10]:
```
np.int64(6501)
```

In [11]:
```
df['Descriptor'].describe()
```

Out[11]:
```
count        358057
unique           45
top       No Access
freq          75888
Name: Descriptor, dtype: object
```

In [12]:
```
df.Descriptor.value_counts()
```

Out[12]:
```
Descriptor
No Access                          75888
Loud Music/Party                   69708
Posted Parking Sign Violation      27200
Partial Access                     24993
Loud Talking                       23810
With License Plate                 21661
Blocked Hydrant                    19989
Commercial Overnight Parking       14997
Blocked Sidewalk                   13860
Car/Truck Music                    11940
Double Parked Blocking Traffic      7164
Engine Idling                       5323
Double Parked Blocking Vehicle      5167
Neglected                           5131
Banging/Pounding                    4959
Car/Truck Horn                      4478
Congestion/Gridlock                 3237
Other (complaint details)           2489
In Prohibited Area                  2206
Overnight Commercial Storage        2102
Unlicensed                          1986
Unauthorized Bus Layover            1618
Truck Route Violation               1125
Tortured                            1041
In Public                            971
No Shelter                           886
Chained                              695
Vehicle                              602
Detached Trailer                     582
Underage - Licensed Est              345
Chronic Stoplight Violation          339
Chronic Speeding                     306
In Car                               299
Playing in Unsuitable Place          272
Drag Racing                          191
Police Report Requested              122
Loud Television                      114
After Hours - Licensed Est            93
Building                              79
Nuisance/Truant                       43
Police Report Not Requested           35
Language Access Complaint              8
Homeless Issue                         1
Disruptive Passenger                   1
Animal Waste                           1
Name: count, dtype: int64
```

In [13]:
```
df['Descriptor'].fillna('No Description', inplace = True)
```

In [14]:
```
df['Descriptor'].isnull().sum()
```

Out[14]:
```
np.int64(0)
```

In [15]:
```
df.Descriptor.value_counts()
```

Out[15]:
```
Descriptor
No Access                          75888
Loud Music/Party                   69708
Posted Parking Sign Violation      27200
Partial Access                     24993
Loud Talking                       23810
With License Plate                 21661
Blocked Hydrant                    19989
Commercial Overnight Parking       14997
Blocked Sidewalk                   13860
Car/Truck Music                    11940
Double Parked Blocking Traffic      7164
No Description                      6501
Engine Idling                       5323
Double Parked Blocking Vehicle      5167
Neglected                           5131
Banging/Pounding                    4959
Car/Truck Horn                      4478
Congestion/Gridlock                 3237
Other (complaint details)           2489
In Prohibited Area                  2206
Overnight Commercial Storage        2102
Unlicensed                          1986
Unauthorized Bus Layover            1618
Truck Route Violation               1125
Tortured                            1041
In Public                            971
No Shelter                           886
Chained                              695
Vehicle                              602
Detached Trailer                     582
Underage - Licensed Est              345
Chronic Stoplight Violation          339
Chronic Speeding                     306
In Car                               299
Playing in Unsuitable Place          272
Drag Racing                          191
Police Report Requested              122
Loud Television                      114
After Hours - Licensed Est            93
Building                              79
Nuisance/Truant                       43
Police Report Not Requested           35
Language Access Complaint              8
Homeless Issue                         1
Disruptive Passenger                   1
Animal Waste                           1
Name: count, dtype: int64
```

Filling NA Value of Location Type Column

In [16]:
```
df['Location Type'].isnull().sum()
```

Out[16]: np.int64(133)

In [17]:
```
df['Location Type'].describe()
```

Out[17]:
```
count              364425
unique                 18
top       Street/Sidewalk
freq               301372
Name: Location Type, dtype: object
```

In [18]:
```
df['Location Type'].value_counts()
```

Out[18]:
```
Location Type
Street/Sidewalk               301372
Store/Commercial               25217
Club/Bar/Restaurant            21527
Residential Building/House      8869
Park/Playground                 4871
House of Worship                1068
Residential Building             533
Highway                          265
House and Store                  245
Parking Lot                      167
Commercial                       108
Vacant Lot                        95
Roadway Tunnel                    43
Subway Station                    40
Bridge                             2
Terminal                           1
Ferry                              1
Park                               1
Name: count, dtype: int64
```

In [19]:
```
df['Location Type'].fillna('Unknown', inplace = True)
```

In [20]:
```
df['Location Type'].isnull().sum()
```

Out[20]: np.int64(0)

In [21]:
```
df['Location Type'].value_counts()
```

Out[21]:
```
Location Type
Street/Sidewalk                301372
Store/Commercial                25217
Club/Bar/Restaurant             21527
Residential Building/House       8869
Park/Playground                  4871
House of Worship                 1068
Residential Building              533
Highway                           265
House and Store                   245
Parking Lot                       167
Unknown                           133
Commercial                        108
Vacant Lot                         95
Roadway Tunnel                     43
Subway Station                     40
Bridge                              2
Terminal                            1
Ferry                               1
Park                                1
Name: count, dtype: int64
```

Filling NA Value of Incident Zip Column

In [22]:
```
df['Incident Zip'].isnull().sum()
```

Out[22]:
```
np.int64(2998)
```

In [23]:
```
df['Incident Zip'].value_counts().head()
```

Out[23]:
```
Incident Zip
11385.0    6396
11368.0    5384
11234.0    5199
11211.0    4854
11377.0    4525
Name: count, dtype: int64
```

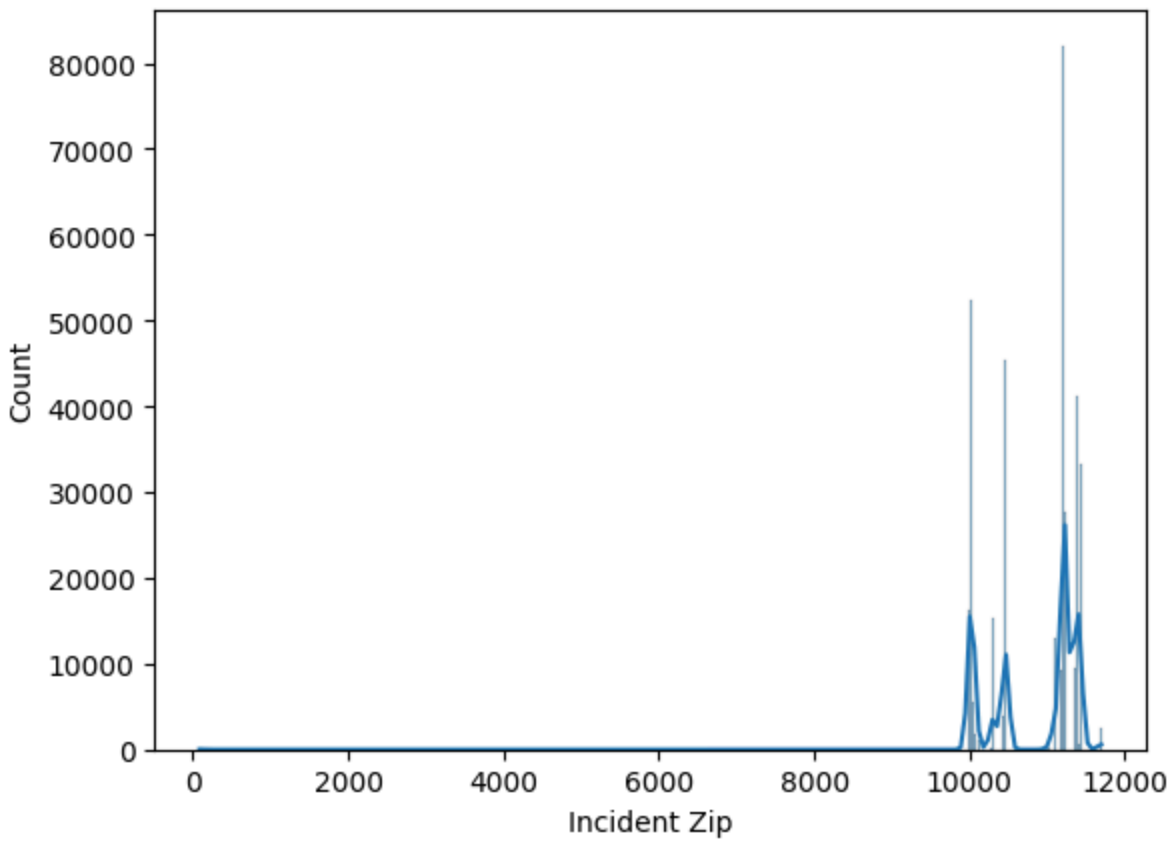
In [24]:
```
df['Incident Zip'].describe()
```

Out[24]:
```
count    361560.000000
mean      10858.496659
std         578.263114
min          83.000000
25%       10314.000000
50%       11209.000000
75%       11238.000000
max       11697.000000
Name: Incident Zip, dtype: float64
```

In [25]:
```
df['Incident Zip'].mode()
```

Out[25]:
```
0    11385.0
Name: Incident Zip, dtype: float64
```

In [26]:
```
sns.histplot(df['Incident Zip'], kde = True)
plt.show()
```



In [27]:
```
sns.distplot(df['Incident Zip'])
plt.show()
```

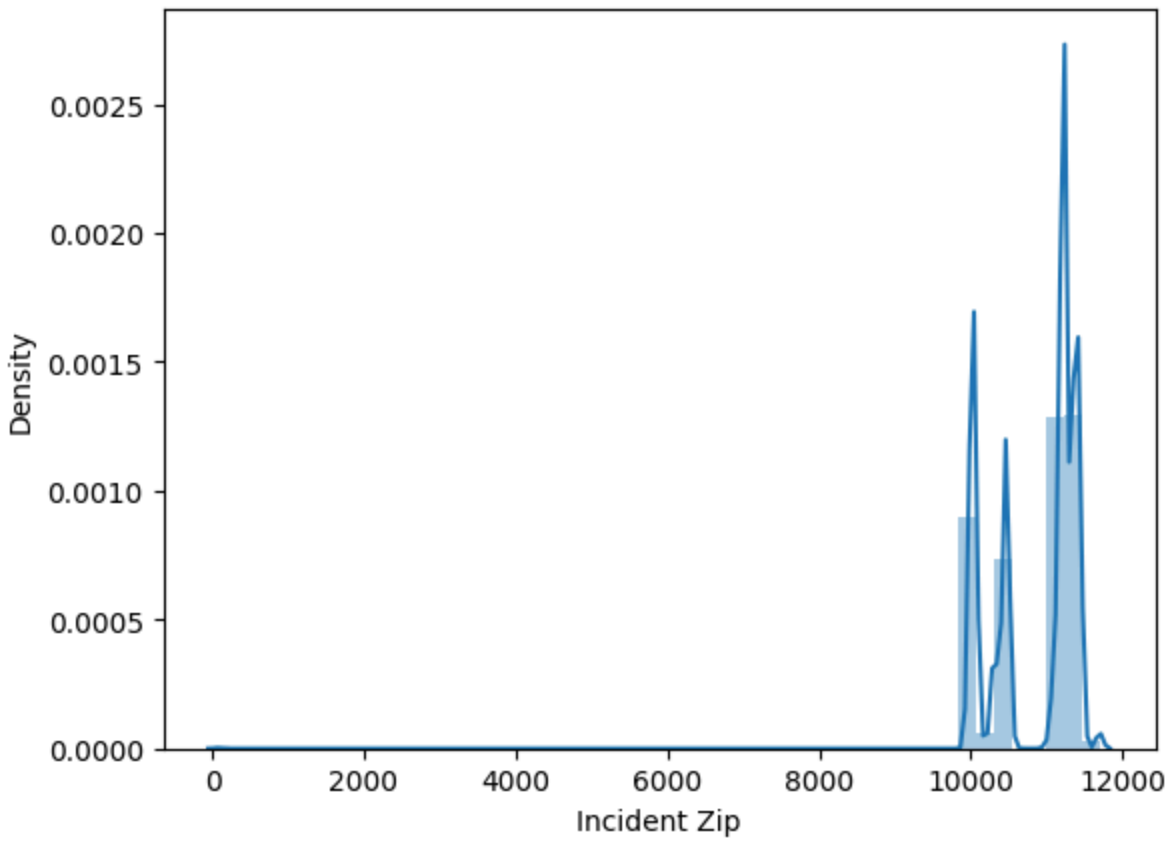
```
C:\Users\DELL\AppData\Local\Temp\ipykernel_14776\3996573223.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['Incident Zip'])
```



In [28]:
```
df['Incident Zip'] = df['Incident Zip'].fillna(df['Incident Zip'].mode()[0])
```

In [29]:
```
df['Incident Zip'].isnull().sum()
```

Out[29]:
```
np.int64(0)
```

In [30]:
```
df['Incident Zip'].value_counts().head()
```

Out[30]:
```
Incident Zip
11385.0    9394
11368.0    5384
11234.0    5199
11211.0    4854
11377.0    4525
Name: count, dtype: int64
```

Study Of Address Type

In [31]:
```
df['Address Type'].isnull().sum()
```

Out[31]:
```
np.int64(3252)
```

In [32]:
```
df['Address Type'].describe()
```

Out[32]:
```
count      361306
unique          5
top       ADDRESS
freq       292699
Name: Address Type, dtype: object
```

In [33]:
```
df['Address Type'].value_counts()
```

Out[33]:
```
Address Type
ADDRESS         292699
INTERSECTION     50514
BLOCKFACE        13786
LATLONG           3931
PLACENAME          376
Name: count, dtype: int64
```

In [34]:
```
df['Address Type'].fillna('Unkonown', inplace = True)
```

In [35]:
```
df['Address Type'].value_counts()
```

Out[35]:
```
Address Type
ADDRESS         292699
INTERSECTION     50514
BLOCKFACE        13786
LATLONG           3931
Unkonown          3252
PLACENAME          376
Name: count, dtype: int64
```

In [36]:
```
df['Address Type'].isnull().sum()
```

Out[36]:
```
np.int64(0)
```

Remove Not Useful Columns Of This Data

In [37]:
```
df.drop(['Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2',], axis = 1, inplace = True
```

In [38]:
```
df.head()
```

Out[38]:

	Unique Key	Created Date	Closed Date	Agency	Complaint Type	Descriptor	Location Type
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	Blocked Driveway	No Access	Street/Sidewalk
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	Blocked Driveway	No Access	Street/Sidewalk
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	Illegal Parking	Commercial Overnight Parking	Street/Sidewalk
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	Illegal Parking	Blocked Sidewalk	Street/Sidewalk

5 rows × 46 columns

Filling NA Value of City Column

In [39]:
```
df['City'].isnull().sum()
```

Out[39]:
```
np.int64(2997)
```

In [40]:
```
df['City'].describe()
```

Out[40]:
```
count       361561
unique          53
top       BROOKLYN
freq        118862
Name: City, dtype: object
```

In [41]:
```
df['City'].value_counts()
```

Out[41]:

```
City
BROOKLYN                118862
NEW YORK                 77312
BRONX                    49171
STATEN ISLAND            15340
JAMAICA                   8932
ASTORIA                   7991
FLUSHING                  7487
RIDGEWOOD                 6392
CORONA                    5383
WOODSIDE                  4357
EAST ELMHURST             3558
OZONE PARK                3446
ELMHURST                  3438
SOUTH RICHMOND HILL       3431
MASPETH                   3118
WOODHAVEN                 3103
LONG ISLAND CITY          3028
SOUTH OZONE PARK          2668
FRESH MEADOWS             2453
RICHMOND HILL             2335
MIDDLE VILLAGE            2291
QUEENS VILLAGE            2251
FOREST HILLS              2122
JACKSON HEIGHTS           2106
REGO PARK                 1807
BAYSIDE                   1550
COLLEGE POINT             1544
FAR ROCKAWAY              1397
WHITESTONE                1369
HOLLIS                    1231
HOWARD BEACH              1144
SPRINGFIELD GARDENS       1094
ROSEDALE                  1091
SAINT ALBANS              1047
KEW GARDENS               1008
SUNNYSIDE                  944
Astoria                    906
ROCKAWAY PARK              831
OAKLAND GARDENS            717
LITTLE NECK                712
CAMBRIA HEIGHTS            617
BELLEROSE                  487
GLEN OAKS                  361
ARVERNE                    259
FLORAL PARK                196
Long Island City           170
Woodside                   166
NEW HYDE PARK              129
CENTRAL PARK               110
QUEENS                      37
BREEZY POINT                31
East Elmhurst               30
Howard Beach                 1
Name: count, dtype: int64
```

In [42]:
```
df['City'].fillna('Unknown City', inplace = True)
```

In [43]:
```
df['City'].isnull().sum()
```

Out[43]:
```
np.int64(0)
```

In [44]:
```
df['City'].value_counts()
```

Out[44]:

```
City
BROOKLYN                118862
NEW YORK                 77312
BRONX                    49171
STATEN ISLAND            15340
JAMAICA                   8932
ASTORIA                   7991
FLUSHING                  7487
RIDGEWOOD                 6392
CORONA                    5383
WOODSIDE                  4357
EAST ELMHURST             3558
OZONE PARK                3446
ELMHURST                  3438
SOUTH RICHMOND HILL       3431
MASPETH                   3118
WOODHAVEN                 3103
LONG ISLAND CITY          3028
Unknown City              2997
SOUTH OZONE PARK          2668
FRESH MEADOWS             2453
RICHMOND HILL             2335
MIDDLE VILLAGE            2291
QUEENS VILLAGE            2251
FOREST HILLS              2122
JACKSON HEIGHTS           2106
REGO PARK                 1807
BAYSIDE                   1550
COLLEGE POINT             1544
FAR ROCKAWAY              1397
WHITESTONE                1369
HOLLIS                    1231
HOWARD BEACH              1144
SPRINGFIELD GARDENS       1094
ROSEDALE                  1091
SAINT ALBANS              1047
KEW GARDENS               1008
SUNNYSIDE                  944
Astoria                    906
ROCKAWAY PARK              831
OAKLAND GARDENS            717
LITTLE NECK                712
CAMBRIA HEIGHTS            617
BELLEROSE                  487
GLEN OAKS                  361
ARVERNE                    259
FLORAL PARK                196
Long Island City           170
Woodside                   166
NEW HYDE PARK              129
CENTRAL PARK               110
QUEENS                      37
BREEZY POINT                31
East Elmhurst               30
Howard Beach                 1
Name: count, dtype: int64
```

Remove Landmark Column

In [45]:
```
df['Landmark'].isnull().sum()
```

Out[45]:
```
np.int64(364183)
```

In [46]:
```
df.drop(['Landmark'], axis = 1, inplace = True)
```

Filling NA Value of Facility Type Column

In [47]:
```
df['Facility Type'].isnull().sum()
```

Out[47]:
```
np.int64(2389)
```

In [48]:
```
df['Facility Type'].describe()
```

Out[48]:
```
count       362169
unique           1
top       Precinct
freq        362169
Name: Facility Type, dtype: object
```

In [49]:
```
df['Facility Type'].fillna('Unknown', inplace = True)
```

In [50]:
```
df['Facility Type'].value_counts()
```

Out[50]:
```
Facility Type
Precinct    362169
Unknown       2389
Name: count, dtype: int64
```

Status Column (Not Changes)

In [51]:
```
df['Status'].isnull().sum()
```

Out[51]:
```
np.int64(0)
```

Remove Due Date Column

In [52]:
```
df.drop(['Due Date'], inplace = True, axis = 1)
```

Resolution Description Column (Not Changes)

In [53]:
```
df['Resolution Description'].isnull().sum()
```

Out[53]:
```
np.int64(0)
```

Study of Resolution Action Updated Date Column

In [54]:
```
df['Resolution Action Updated Date'].isnull().sum()
```

Out[54]:
```
np.int64(2402)
```

In [55]:
```
df['Resolution Action Updated Date'].describe()
```

Out[55]:
```
count                     362156
unique                    340833
top       12/19/2015 08:35:28 AM
freq                           3
Name: Resolution Action Updated Date, dtype: object
```

In [56]:
```
df = df[df['Resolution Action Updated Date'].notnull()]
```

In [57]:
```
df['Resolution Action Updated Date'].isnull().sum()
```

Out[57]:
```
np.int64(0)
```

Remove Borough to Location Columns

In [58]:
```
df.columns
```

Out[58]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Complaint Type',
       'Descriptor', 'Location Type', 'Incident Zip', 'Address Type', 'City',
       'Facility Type', 'Status', 'Resolution Description',
       'Resolution Action Updated Date', 'Community Board', 'Borough',
       'X Coordinate (State Plane)', 'Y Coordinate (State Plane)',
       'Park Facility Name', 'Park Borough', 'School Name', 'School Number',
       'School Region', 'School Code', 'School Phone Number', 'School Address',
       'School City', 'School State', 'School Zip', 'School Not Found',
       'School or Citywide Complaint', 'Vehicle Type', 'Taxi Company Borough',
       'Taxi Pick Up Location', 'Bridge Highway Name',
       'Bridge Highway Direction', 'Road Ramp', 'Bridge Highway Segment',
       'Garage Lot Name', 'Ferry Direction', 'Ferry Terminal Name', 'Latitude',
       'Longitude', 'Location'],
      dtype='object')
```

In [59]:
```
df.drop(['Borough',
       'X Coordinate (State Plane)', 'Y Coordinate (State Plane)',
       'Park Facility Name', 'Park Borough', 'School Name', 'School Number',
       'School Region', 'School Code', 'School Phone Number', 'School Address',
       'School City', 'School State', 'School Zip', 'School Not Found',
       'School or Citywide Complaint'], axis = 1, inplace = True)
```

In [60]:
```
df.drop(['Vehicle Type', 'Taxi Company Borough',
       'Taxi Pick Up Location', 'Bridge Highway Name',
       'Bridge Highway Direction', 'Road Ramp', 'Bridge Highway Segment',
       'Garage Lot Name', 'Ferry Direction', 'Ferry Terminal Name'], axis = 1, inpl
```

In [61]:
```
df['Location'].duplicated().sum()
```

Out[61]:
```
np.int64(215424)
```

In [62]:
```
df.drop(['Location'], axis = 1, inplace = True)
```

Study Of Longitude & Latitude Columns

In [63]:
```
df[['Latitude', 'Longitude']].isnull().sum()
```

Out[63]:
```
Latitude     1708
Longitude    1708
dtype: int64
```

In [64]:
```
df['Latitude'].describe()
```

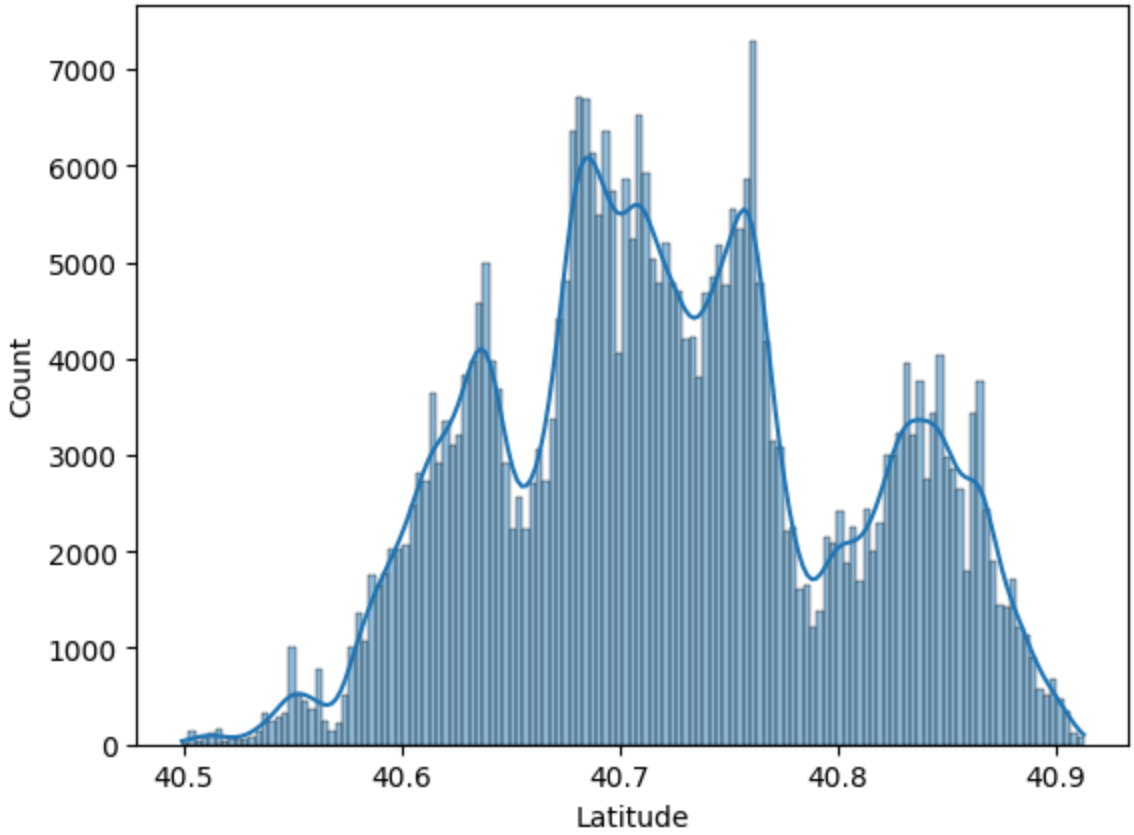
Out[64]:
```
count    360448.000000
mean         40.724980
std           0.081908
min          40.499040
25%          40.668739
50%          40.718410
75%          40.778166
max          40.912869
Name: Latitude, dtype: float64
```

In [65]:
```
df = df[df['Latitude'].notnull()]
df = df[df['Longitude'].notnull()]
```

In [66]:
```
df[['Latitude', 'Longitude']].isnull().sum()
```

Out[66]:
```
Latitude     0
Longitude    0
dtype: int64
```

In [67]:
```
sns.histplot(df['Latitude'], kde = True)
plt.show()
```



Final Check

In [68]:
```
df.columns
```

Out[68]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Complaint Type',
       'Descriptor', 'Location Type', 'Incident Zip', 'Address Type', 'City',
       'Facility Type', 'Status', 'Resolution Description',
       'Resolution Action Updated Date', 'Community Board', 'Latitude',
       'Longitude'],
      dtype='object')
```

In [69]:
```
df.isnull().sum()
```

Out[69]:
```
Unique Key                        0
Created Date                      0
Closed Date                      17
Agency                            0
Complaint Type                    0
Descriptor                        0
Location Type                     0
Incident Zip                      0
Address Type                      0
City                              0
Facility Type                     0
Status                            0
Resolution Description            0
Resolution Action Updated Date    0
Community Board                   0
Latitude                          0
Longitude                         0
dtype: int64
```

In [70]:
```
df.shape
```

Out[70]:
```
(360448, 17)
```

In [71]:
```
df.head()
```

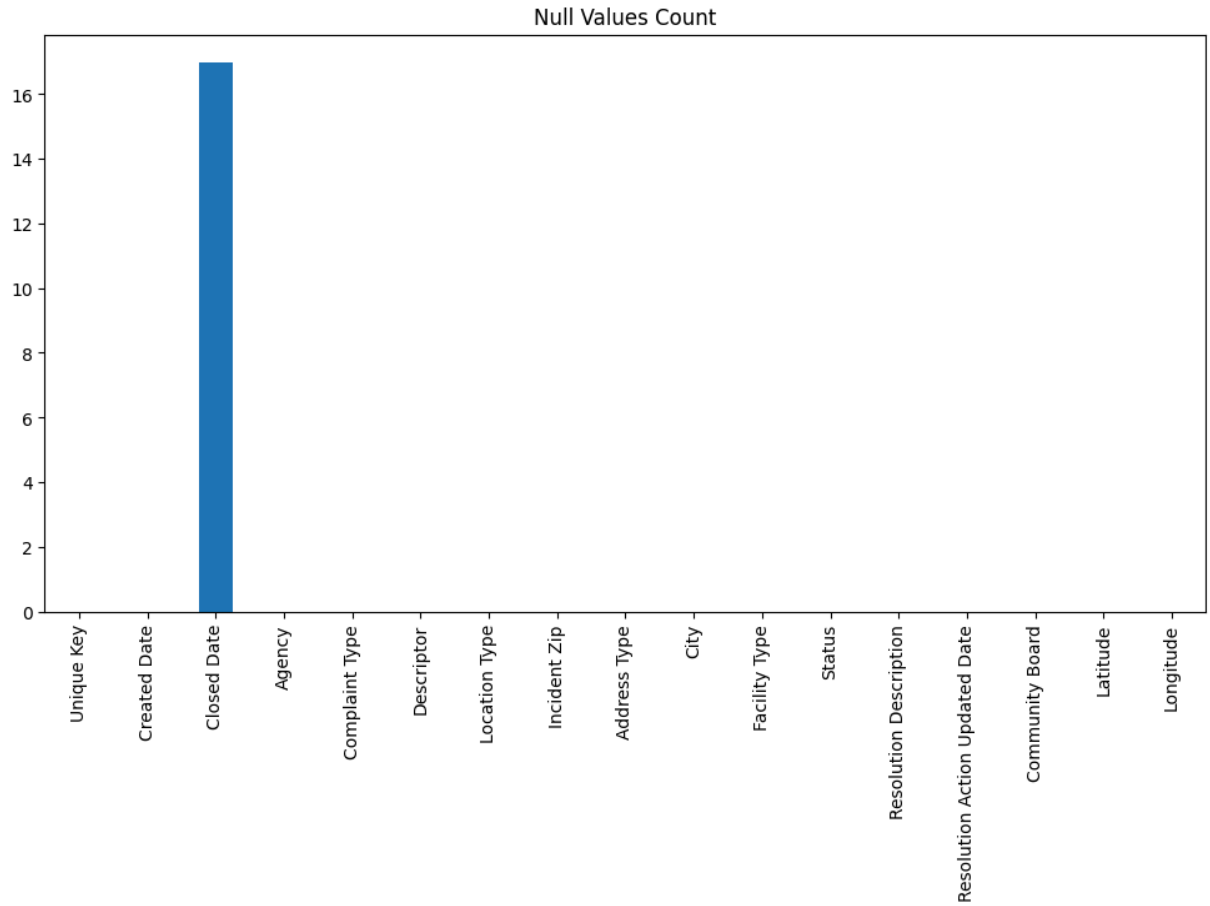
Out[71]:

	Unique Key	Created Date	Closed Date	Agency	Complaint Type	Descriptor	Location Type
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	Blocked Driveway	No Access	Street/Sidewalk
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	Blocked Driveway	No Access	Street/Sidewalk
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	Illegal Parking	Commercial Overnight Parking	Street/Sidewalk
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	Illegal Parking	Blocked Sidewalk	Street/Sidewalk

Step 2 :- Exploratory Data Analysis

Frequency Plot Of Null Values

In [72]:
```
df.isnull().sum().plot(kind = 'bar', figsize = (12, 6))
plt.title('Null Values Count')
plt.show()
```



--Missing Value Treatments

Removes rows where Closed Date is Null

In [74]:
```
df = df[df['Closed Date'].notnull()]
```

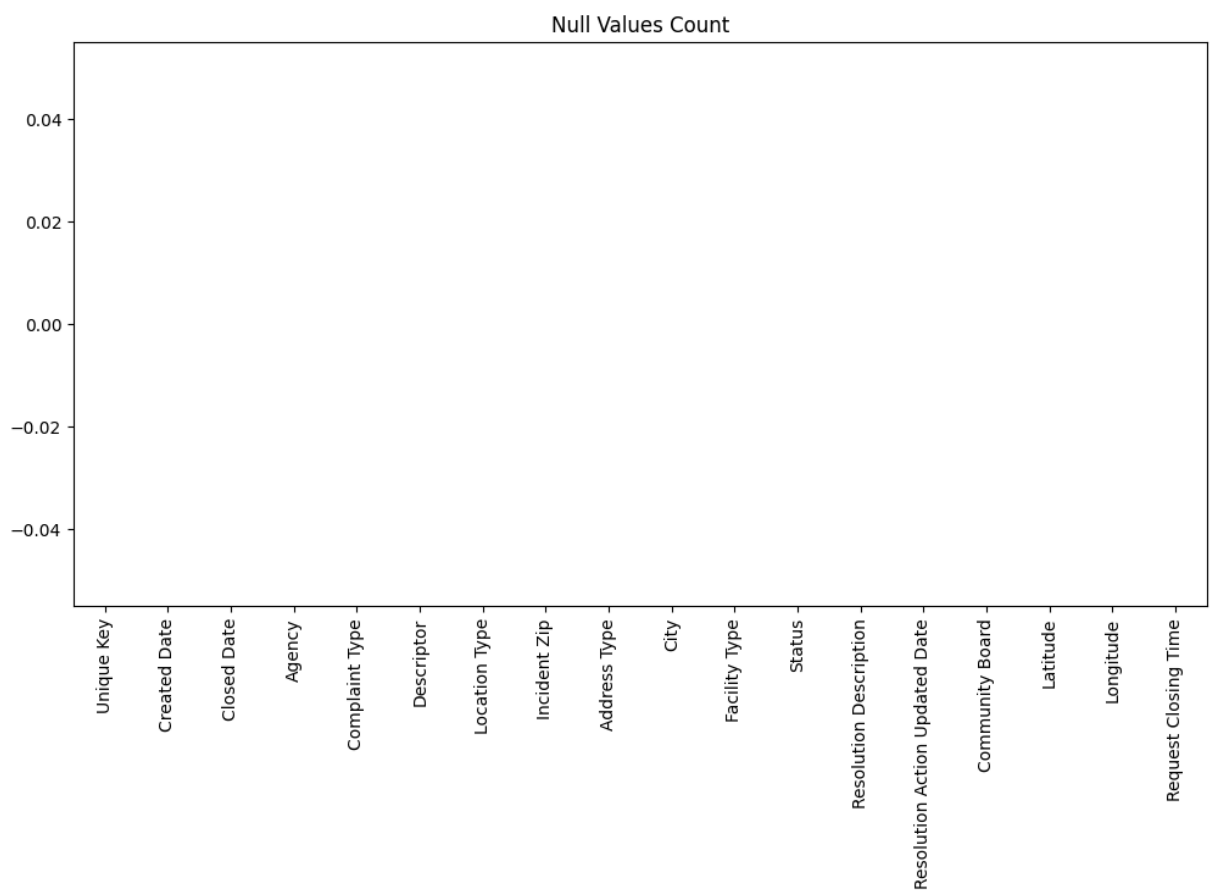
In [75]:
```
df['Closed Date'].isnull().sum()
```

Out[75]:
```
np.int64(0)
```

In [76]:
```
df.isnull().sum()
```

Out[76]:
```
Unique Key                        0
Created Date                      0
Closed Date                       0
Agency                            0
Complaint Type                    0
Descriptor                        0
Location Type                     0
Incident Zip                      0
Address Type                      0
City                              0
Facility Type                     0
Status                            0
Resolution Description            0
Resolution Action Updated Date    0
Community Board                   0
Latitude                          0
Longitude                         0
dtype: int64
```

In [108…
```
df.isnull().sum().plot(kind = 'bar', figsize = (12, 6))
plt.title('Null Values Count')
plt.show()
```



In [ ]:
```
# After data cleaning, there were no missing value left in the dataset,
# so the null value frequency plot appear empty, indicating successfully preprocess
```

Analyze the Date Column

In [77]:
```
df = df.copy()

df['Created Date'] = pd.to_datetime(df['Created Date'], errors='coerce')
df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
```

```
C:\Users\DELL\AppData\Local\Temp\ipykernel_14776\1077711164.py:3: UserWarning: Could
not infer format, so each element will be parsed individually, falling back to `date
util`. To ensure parsing is consistent and as-expected, please specify a format.
  df['Created Date'] = pd.to_datetime(df['Created Date'], errors='coerce')
C:\Users\DELL\AppData\Local\Temp\ipykernel_14776\1077711164.py:4: UserWarning: Could
not infer format, so each element will be parsed individually, falling back to `date
util`. To ensure parsing is consistent and as-expected, please specify a format.
  df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
```

In [78]:
```
df[['Created Date', 'Closed Date']].dtypes
```

Out[78]:
```
Created Date    datetime64[ns]
Closed Date     datetime64[ns]
dtype: object
```

In [79]:
```
df[['Created Date', 'Closed Date']].head()
```

Out[79]:

	Created Date	Closed Date
0	2015-12-31 23:59:45	2016-01-01 00:55:15
1	2015-12-31 23:59:44	2016-01-01 01:26:57
2	2015-12-31 23:59:29	2016-01-01 04:51:03
3	2015-12-31 23:57:46	2016-01-01 07:43:13
4	2015-12-31 23:56:58	2016-01-01 03:24:42

Remove Incorrect Timeline

In [80]:
```
df = df[df['Closed Date'] >= df['Created Date']]
```

In [81]:
```
df[['Created Date', 'Closed Date']].head()
```

Out[81]:

	Created Date	Closed Date
0	2015-12-31 23:59:45	2016-01-01 00:55:15
1	2015-12-31 23:59:44	2016-01-01 01:26:57
2	2015-12-31 23:59:29	2016-01-01 04:51:03
3	2015-12-31 23:57:46	2016-01-01 07:43:13
4	2015-12-31 23:56:58	2016-01-01 03:24:42

Calculate Time Elapsed in Closed & Creation Date

In [83]:
```
df['Request Closing Time'] = df['Closed Date'] - df['Created Date']
```

In [84]:
```
df.shape
```

Out[84]:
```
(360431, 18)
```

In [85]:
```
df.columns
```

Out[85]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Complaint Type',
       'Descriptor', 'Location Type', 'Incident Zip', 'Address Type', 'City',
       'Facility Type', 'Status', 'Resolution Description',
       'Resolution Action Updated Date', 'Community Board', 'Latitude',
       'Longitude', 'Request Closing Time'],
      dtype='object')
```

Convert Calculated Date to Seconds

In [86]:
```
df['Request Closing Time'] = df['Request Closing Time'].dt.total_seconds()
```

In [87]:
```
df['Request Closing Time'].dtype
```

Out[87]:
```
dtype('float64')
```

View descriptive statistics for the newly created column

In [88]:
```
df['Request Closing Time'].describe()
```

Out[88]:
```
count    3.604310e+05
mean     1.508083e+04
std      2.097421e+04
min      6.100000e+01
25%      4.529000e+03
50%      9.604000e+03
75%      1.884100e+04
max      2.134342e+06
Name: Request Closing Time, dtype: float64
```

Check null values of Complaint Type & City

In [89]:
```
df[['Complaint Type', 'City']].isnull().sum()
```

Out[89]:
```
Complaint Type    0
City              0
dtype: int64
```

Filling NA Value with Unknown City in City Column

In [90]:
```
# df['City'].fillna('Unknown City, inplace = True')
```

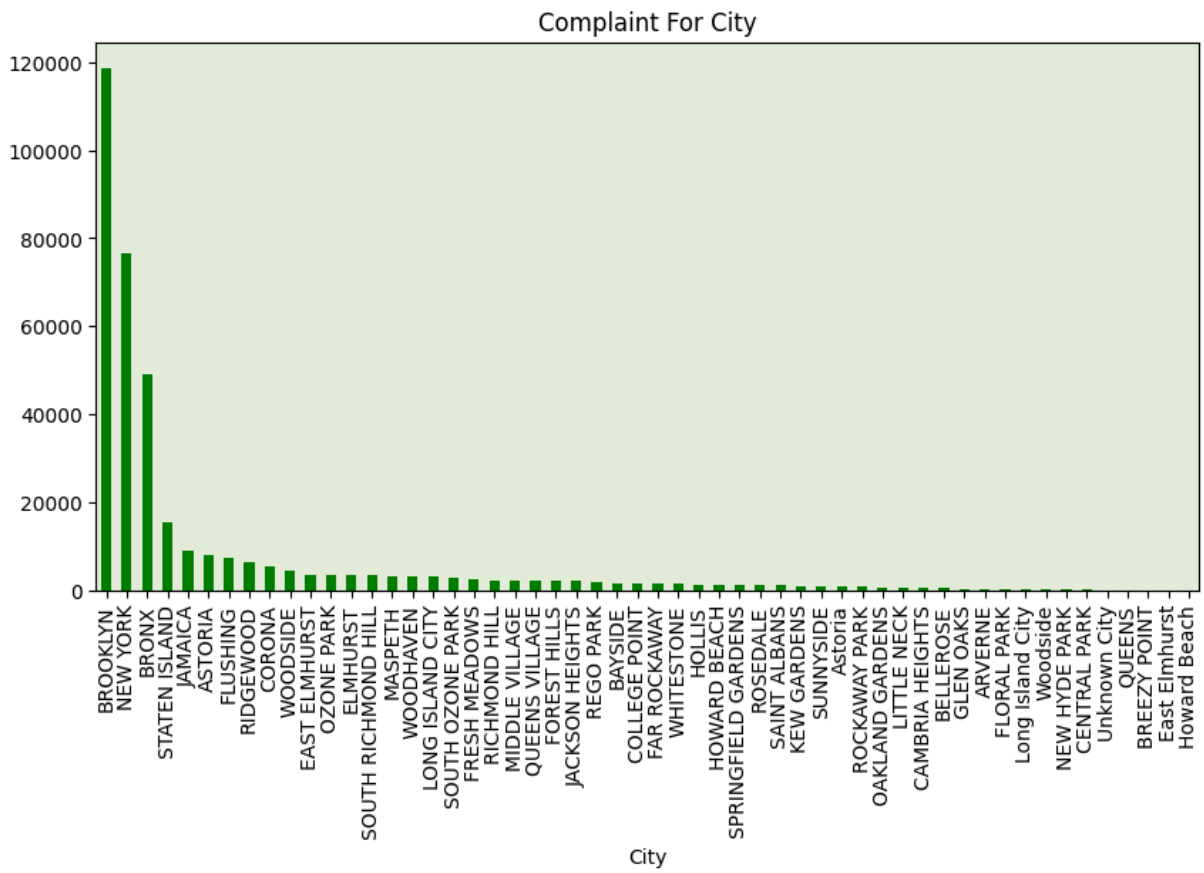
In [91]:
```
df['City'].value_counts()
```

Out[91]:

```
City
BROOKLYN                118624
NEW YORK                 76626
BRONX                    49045
STATEN ISLAND            15323
JAMAICA                   8911
ASTORIA                   7974
FLUSHING                  7481
RIDGEWOOD                 6388
CORONA                    5381
WOODSIDE                  4353
EAST ELMHURST             3557
OZONE PARK                3446
ELMHURST                  3437
SOUTH RICHMOND HILL       3430
MASPETH                   3116
WOODHAVEN                 3101
LONG ISLAND CITY          3018
SOUTH OZONE PARK          2668
FRESH MEADOWS             2447
RICHMOND HILL             2333
MIDDLE VILLAGE            2290
QUEENS VILLAGE            2251
FOREST HILLS              2120
JACKSON HEIGHTS           2105
REGO PARK                 1805
BAYSIDE                   1548
COLLEGE POINT             1544
FAR ROCKAWAY              1396
WHITESTONE                1367
HOLLIS                    1231
HOWARD BEACH              1143
SPRINGFIELD GARDENS       1093
ROSEDALE                  1086
SAINT ALBANS              1047
KEW GARDENS               1008
SUNNYSIDE                  944
Astoria                    905
ROCKAWAY PARK              829
OAKLAND GARDENS            715
LITTLE NECK                712
CAMBRIA HEIGHTS            617
BELLEROSE                  487
GLEN OAKS                  361
ARVERNE                    258
FLORAL PARK                196
Long Island City           170
Woodside                   166
NEW HYDE PARK              129
CENTRAL PARK               110
Unknown City                41
QUEENS                      36
BREEZY POINT                31
East Elmhurst               30
Howard Beach                 1
Name: count, dtype: int64
```

Draw frequency plot for the complaint in each city

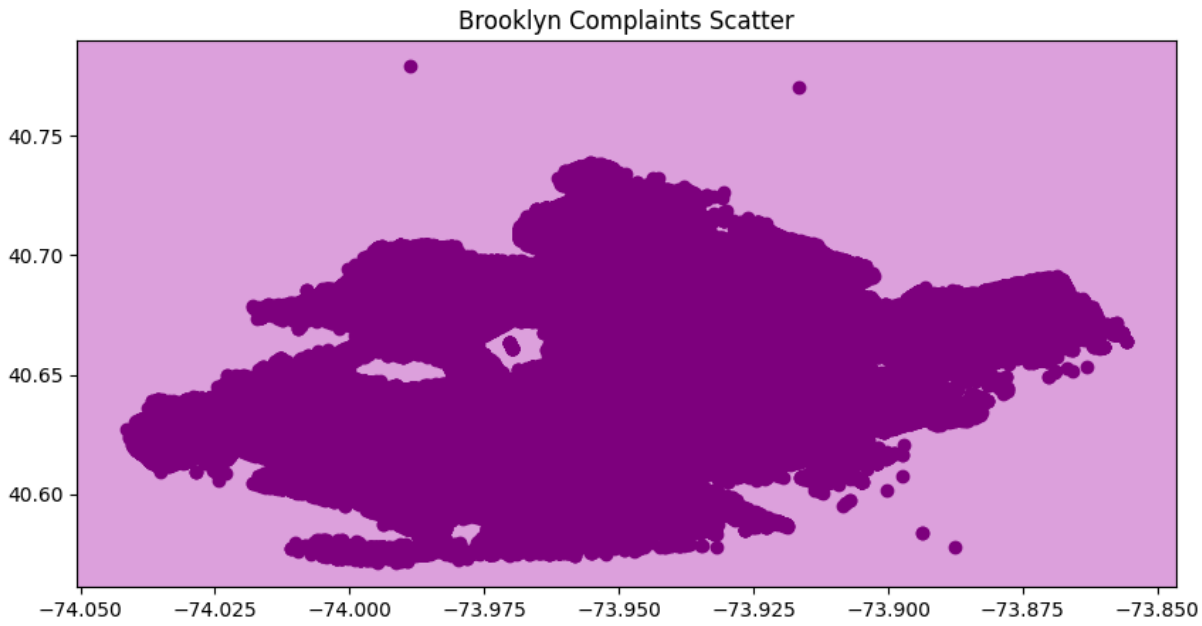
In [92]:
```
plt.figure(figsize = (10,5))
ax = df['City'].value_counts().plot(kind = 'bar', color = 'green')
ax.set_facecolor('#E2EFD9')
plt.title('Complaint For City')
plt.show()
```



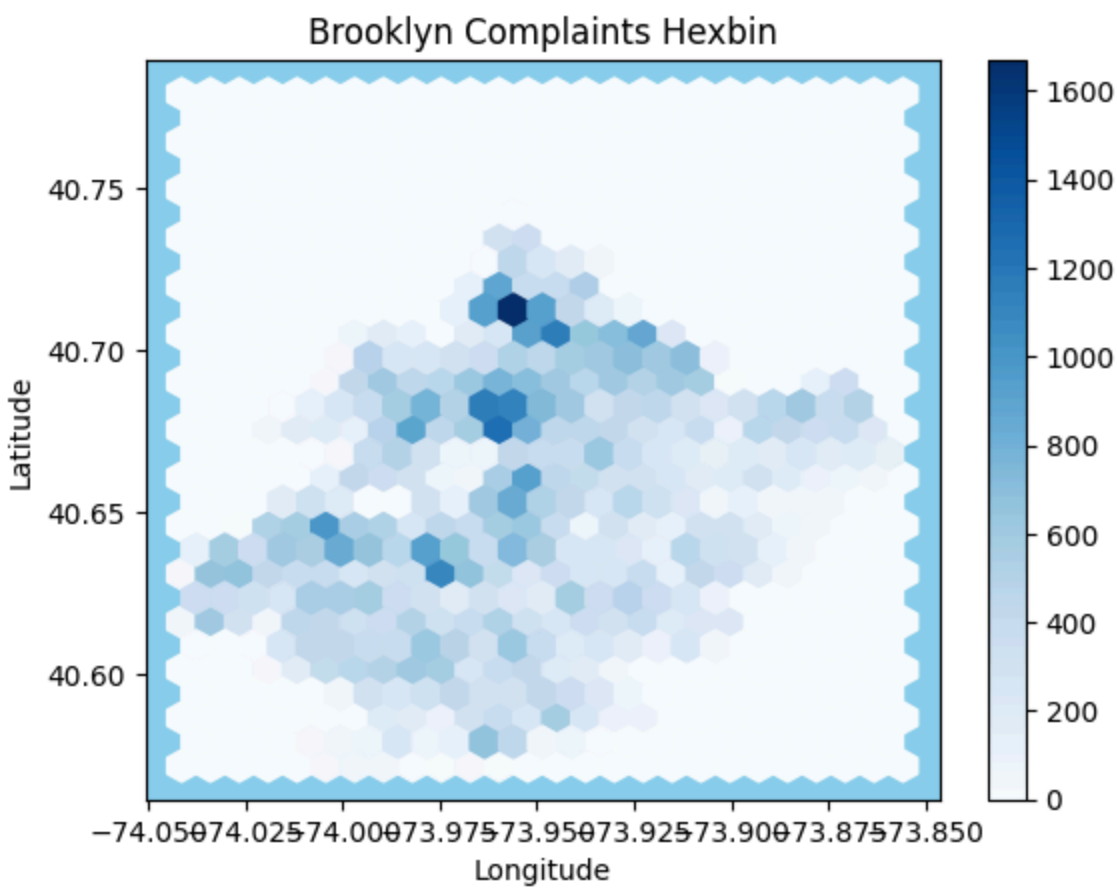
Create a Scatter & Hexbin Plot of the concentration of complaints across Brooklyn

In [93]:
```
plt.figure(figsize = (10,5))
brooklyn = df[df['City'] == "BROOKLYN"]
plt.scatter(x = brooklyn['Longitude'], y = brooklyn['Latitude'], color = '#800080')
plt.gca().set_facecolor('#DDA0DD')
# ax.set_facecolor('#EE82EE')
plt.title('Brooklyn Complaints Scatter')
plt.show()
plt.figure(figsize = (10,5))
ax = brooklyn.plot.hexbin(x = 'Longitude', y = 'Latitude', gridsize = 25, cmap ='Bl
```

```
ax.set_facecolor('#87CEEB')
plt.title('Brooklyn Complaints Hexbin')
plt.show()
```




```
<Figure size 1000x500 with 0 Axes>
```



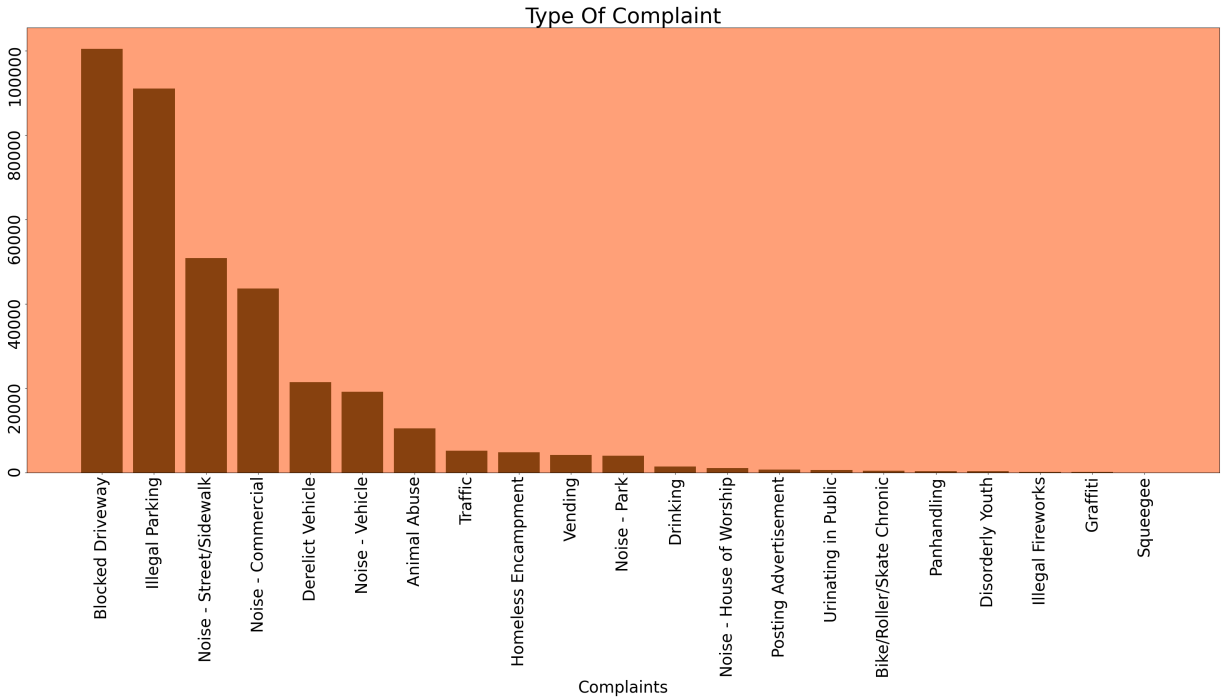
Step 3 :- Major Complaints Types

Plot a Bar Graph to show the Type of Complaint

In [94]:
```
com_type = df['Complaint Type'].value_counts()
print(type(com_type))
print(com_type)
```

```
<class 'pandas.core.series.Series'>
Complaint Type
Blocked Driveway             100487
Illegal Parking               91084
Noise - Street/Sidewalk       50815
Noise - Commercial            43638
Derelict Vehicle              21425
Noise - Vehicle               19124
Animal Abuse                  10503
Traffic                        5166
Homeless Encampment            4828
Vending                        4162
Noise - Park                   3994
Drinking                       1400
Noise - House of Worship       1061
Posting Advertisement           678
Urinating in Public             641
Bike/Roller/Skate Chronic       463
Panhandling                     320
Disorderly Youth                314
Illegal Fireworks               167
Graffiti                        157
Squeegee                          4
Name: count, dtype: int64
```

In [113…
```
plt.figure(figsize = (40,15))
plt.bar(x = com_type.index, height = com_type, color = '#8B4513')
plt.title('Type Of Complaint', fontsize = 40)
plt.xlabel('Complaints', fontsize = 30)
plt.xticks(rotation = 90, size = 30)
plt.yticks(rotation = 90, size = 30)
plt.gca().set_facecolor('#FFA07A')
plt.show()
```



Check the frequency of various types of complaint for New York City

In [96]:
```
nyc = df[df['City'] == 'NEW YORK']
nyc['Complaint Type'].value_counts()
```

Out[96]:
```
Complaint Type
Noise - Street/Sidewalk      22079
Noise - Commercial           18668
Illegal Parking              14366
Noise - Vehicle               6179
Homeless Encampment           3020
Blocked Driveway              2687
Vending                       2619
Animal Abuse                  1926
Traffic                       1749
Noise - Park                  1200
Derelict Vehicle               688
Drinking                       320
Urinating in Public            264
Bike/Roller/Skate Chronic      249
Noise - House of Worship       217
Panhandling                    203
Disorderly Youth                80
Posting Advertisement           49
Illegal Fireworks               34
Graffiti                        25
Squeegee                         4
Name: count, dtype: int64
```

Top 10 Complaints Types

In [97]:
```
df['Complaint Type'].value_counts().head(10)
```

Out[97]:
```
Complaint Type
Blocked Driveway           100487
Illegal Parking             91084
Noise - Street/Sidewalk     50815
Noise - Commercial          43638
Derelict Vehicle            21425
Noise - Vehicle             19124
Animal Abuse                10503
Traffic                      5166
Homeless Encampment          4828
Vending                      4162
Name: count, dtype: int64
```

Display the various type of complaints in each city

In [98]:
```
df.groupby('City')['Complaint Type'].value_counts()
```

Out[98]:
```
City      Complaint Type
ARVERNE   Illegal Parking             62
          Blocked Driveway            50
          Animal Abuse                46
          Derelict Vehicle            32
          Noise - Street/Sidewalk     29
                                     ...
Woodside  Illegal Parking            124
          Blocked Driveway            27
          Derelict Vehicle             8
          Noise - Street/Sidewalk      5
          Noise - Commercial           2
Name: count, Length: 782, dtype: int64
```

Create a Dataframe, df_new (Create Pivot Table)

In [99]:
```
df_new = pd.crosstab(df['Complaint Type'], df['City'])
df_new.head()
```

Out[99]:

City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX	BI
Complaint Type								
Animal Abuse	46	170	0	53	15	2	1967	
Bike/Roller/Skate Chronic	0	16	0	0	1	0	22	
Blocked Driveway	50	3436	159	514	138	3	17051	
Derelict Vehicle	32	426	14	231	120	3	2399	
Disorderly Youth	2	5	0	2	2	0	66	

5 rows × 54 columns

Step 4 :- Visualization

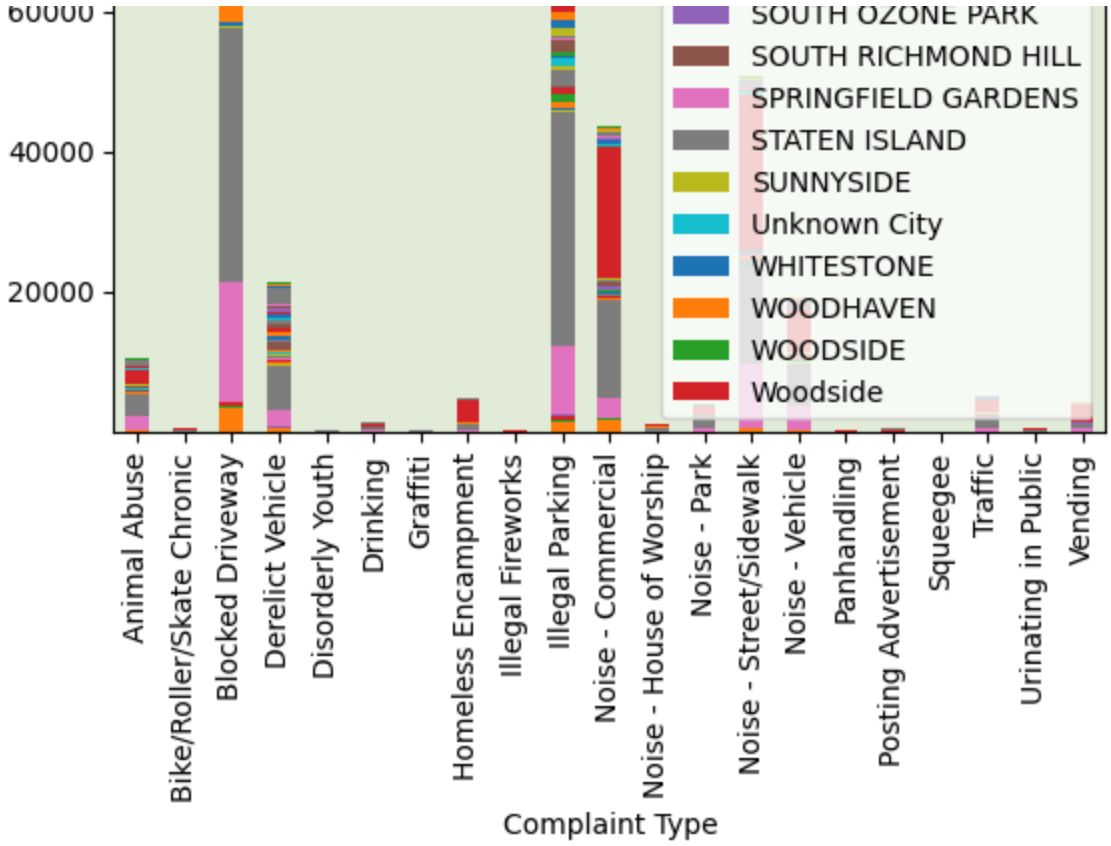
Multi City Complaint Chart

In [117…

```
plt.figure(figsize = (10, 5))
ax = df_new.plot(kind = 'bar', stacked = True)
plt.title('Multi City Complaint Types')
ax.set_facecolor('#E2EFD9')
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

City
ARVERNE
ASTORIA
Astoria
BAYSIDE
BELLEROSE
BREEZY POINT
BRONX
BROOKLYN
CAMBRIA HEIGHTS
CENTRAL PARK
COLLEGE POINT
CORONA
EAST ELMHURST
ELMHURST
East Elmhurst
FAR ROCKAWAY
FLORAL PARK
FLUSHING
FOREST HILLS
FRESH MEADOWS
GLEN OAKS
HOLLIS
HOWARD BEACH
Howard Beach
JACKSON HEIGHTS
JAMAICA
KEW GARDENS
LITTLE NECK
LONG ISLAND CITY
Long Island City
MASPETH
MIDDLE VILLAGE
NEW HYDE PARK
NEW YORK
OAKLAND GARDENS
OZONE PARK
QUEENS
QUEENS VILLAGE
REGO PARK
RICHMOND HILL
RIDGEWOOD
ROCKAWAY PARK
ROSEDALE
SAINT ALBANS
Multi City Complaint Types
100000
80000



Sort by average Request Closing Timegrouping

In [101…

```
df.groupby('Complaint Type')['Request Closing Time'].mean().sort_values()
```

Out[101…

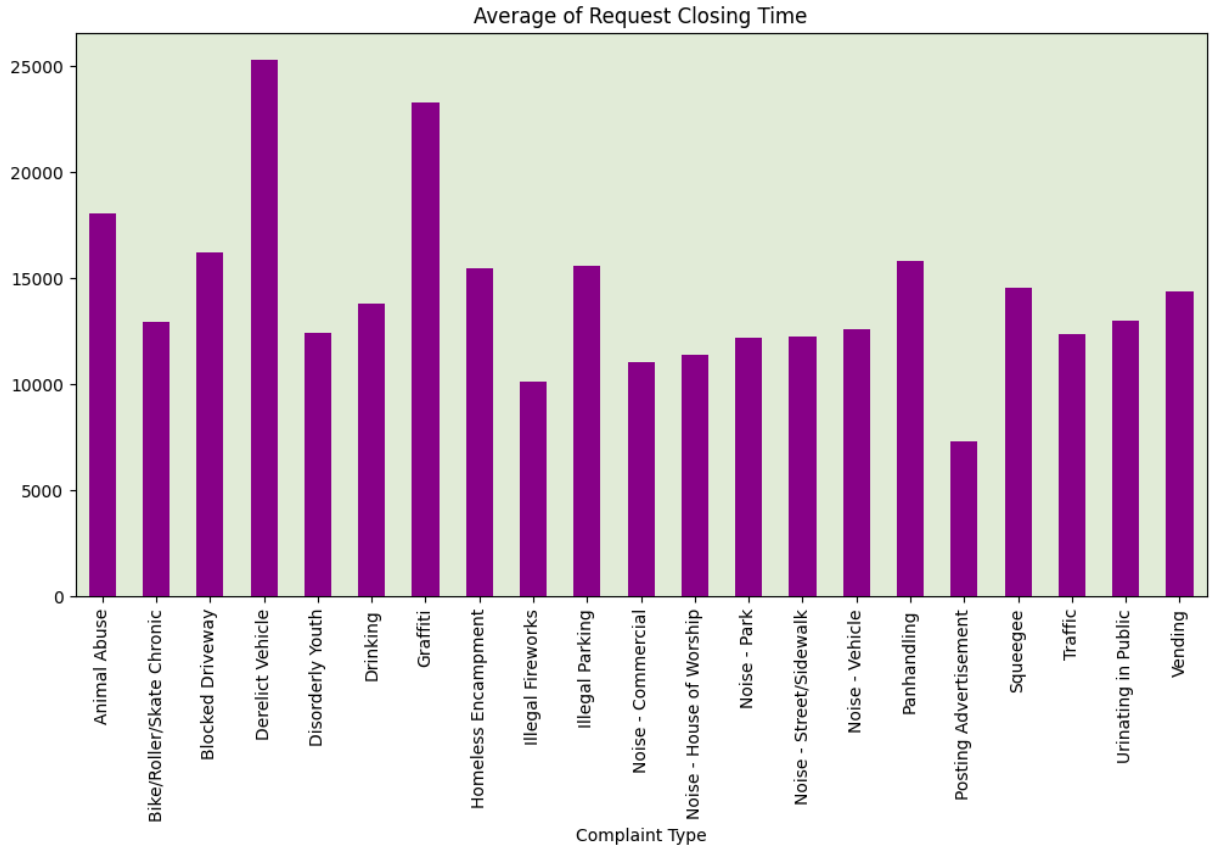
```
Complaint Type
Posting Advertisement         7287.171091
Illegal Fireworks            10122.347305
Noise - Commercial           11040.838260
Noise - House of Worship     11398.217719
Noise - Park                 12196.250125
Noise - Street/Sidewalk      12218.206435
Traffic                      12326.666280
Disorderly Youth             12395.754777
Noise - Vehicle              12586.443840
Bike/Roller/Skate Chronic    12941.995680
Urinating in Public          12959.293292
Drinking                     13803.985000
Vending                      14355.265738
Squeegee                     14560.250000
Homeless Encampment          15479.961268
Illegal Parking              15583.577774
Panhandling                  15822.803125
Blocked Driveway             16217.725089
Animal Abuse                 18051.110921
Graffiti                     23276.343949
Derelict Vehicle             25281.671972
Name: Request Closing Time, dtype: float64
```

Step 5 :- Average Response Time

Visualize the Average of Request Closing Time

In [118…

```
plt.figure(figsize = (12,6))
ax = df.groupby('Complaint Type')['Request Closing Time'].mean().plot(kind = 'bar',
ax.set_facecolor('#E2EFD9')
plt.title('Average of Request Closing Time')
plt.show()
```



Step 6 :- Statistical Analysis using P - Values

In [103…

```
from scipy.stats import f_oneway

groups = [group['Request Closing Time'].values
          for name, group in df.groupby('Complaint Type')]

f_stat, p_value = f_oneway(*groups)

print(p_value)
```

```
0.0
```

Step 7 :- Kruskal-Wallis Test

In [104…

```
from scipy.stats import kruskal

groups = [group['Request Closing Time'].values
          for name, group in df.groupby('Complaint Type')]

stat, p = kruskal(*groups)

print(p)
```

```
0.0
```

Step 8 :- Present Observation

✔️ Missing data handled successfully

✔️ Most complaints come from specific cities

✔️ Some complaint types take longer time

✔️ Response time varies significantly

In [105…

```
df.to_csv('Customer Service Analysis.csv', index = False)
print('Saved Successfully')
```

```
Saved Successfully
```